**BVS1 · Winter Semester 2026/2027 · TH Köln · ICCT**

# Handout — Development Environment

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This handout collects commands that are useful after the initial Lab 0 setup. Use Lab 0 for the linear setup path; use this sheet for everyday operation, backup, and recovery.

## Everyday VM Commands

Run these on the physical host:

```
multipass shell bvs1
multipass stop bvs1
multipass start bvs1
multipass info bvs1
```

Inside the VM, the course repository is:

```
cd ~/teaching-bvs1-code
```

## Keep the VM Updated

Update the VM packages regularly, for example once a week. Inside the VM:

```
sudo apt update
sudo apt upgrade
```

If a restart is required, run `multipass restart bvs1` on the physical host. Afterwards, run `make check` in `~/teaching-bvs1-code`. Stay on Ubuntu 26.04 LTS: do not run `do-release-upgrade`. Avoid updating immediately before a checkpoint appointment.

## Back Up Your Work

Local Git history is not a backup because the repository is stored inside the `bvs1` VM.

If you already use Git and GitHub, you may fork the public course repository, clone your fork inside the VM, and use the usual commit/push workflow to keep your work in your own remote repository. This is optional; the course does not require a GitHub account or a fork.

If you do not want to use GitHub, create a Git bundle and copy it out of the VM.

Inside the VM:

```
cd ~/teaching-bvs1-code
git bundle create ~/teaching-bvs1-backup.bundle --all
```

Return to the physical host:

```
exit
```

Copy the bundle out of the VM:

```
multipass transfer \
  bvs1:/home/ubuntu/teaching-bvs1-backup.bundle .
```

On Windows, Multipass may then print an error similar to:

```
[sftp] cannot set permissions for local file
.\teaching-bvs1-backup.bundle
```

This message concerns setting Unix-style file permissions on the Windows destination. If the bundle exists on the physical host and has a non-zero size, the transfer itself succeeded. In PowerShell you can check:

```
Get-Item .\teaching-bvs1-backup.bundle
```

If the file is shown and has a non-zero size, the transfer succeeded. Keep the bundle outside the VM, preferably in a location that is backed up regularly, such as your Sciebo folder or another backup location.

## Troubleshooting

Stop at the first broken layer. Do not continue with later commands on the wrong machine.

| Problem | Check |
|---|---|
| Multipass cannot read `bvs1-cloud-init.yaml` | Run `pwd` and `ls -l bvs1-cloud-init.yaml*` on the physical host. If the file ends in `.sec` (added by ILU), rename it to `bvs1-cloud-init.yaml`. |
| `multipass shell bvs1` says the instance does not exist | Create the VM first. Do not continue with Git or build commands on the physical host. |
| `cloud-init status --wait` reports an error | Run `cloud-init status --long` and `sudo tail -n 100 /var/log/cloud-init-output.log` inside the VM. |
| `make check` reports a missing tool | Verify `cloud-init status --wait` and `hostname`. Do not install arbitrary replacement packages. |
| `Hello BVS1` appears but no prompt returns | This is expected. Terminate QEMU with Ctrl-a, release, then x. |

### Recreate the VM During Initial Setup

If provisioning fails during Lab 0 before you have started course work, you may recreate the VM from the physical host:

```
multipass delete --purge bvs1
```

Then repeat the Lab 0 launch step.

You may also recreate the VM later in the semester if necessary. Before deleting it, back up the course repository stored inside the VM; `multipass delete --purge bvs1` permanently removes that local repository together with the VM.